CENG 463 – Machine Learning

Take-Home Midterm Examination

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1. Question 1 – Regression: Regularisation, Feature Engineering, and Model Selection

1.1 Problem Formulation and Methodology

The objective of this study is to rigorously evaluate regression architectures on the California Housing dataset, focusing on the impact of advanced regularisation, feature engineering, and statistical validation. The target variable represents median house values, which exhibits a right-skewed distribution. Consequently, a logarithmic transformation ($\log(1 + y)$) was applied to the target variable to stabilise variance and align with the normality assumptions of linear models.

1.2 Experimental Setup and Feature Engineering

A strict hold-out test set (20%) was sequestered prior to any analysis to ensure unbiased final evaluation.

1.2.1 Exploratory Data Analysis (EDA) and Preprocessing

EDA revealed significant variations in feature scales and right-skewed distributions across variables such as *Population* and *MedInc*. Outliers were identified and removed strictly from the training set using the Interquartile Range (IQR) method to prevent model distortion while avoiding data leakage.

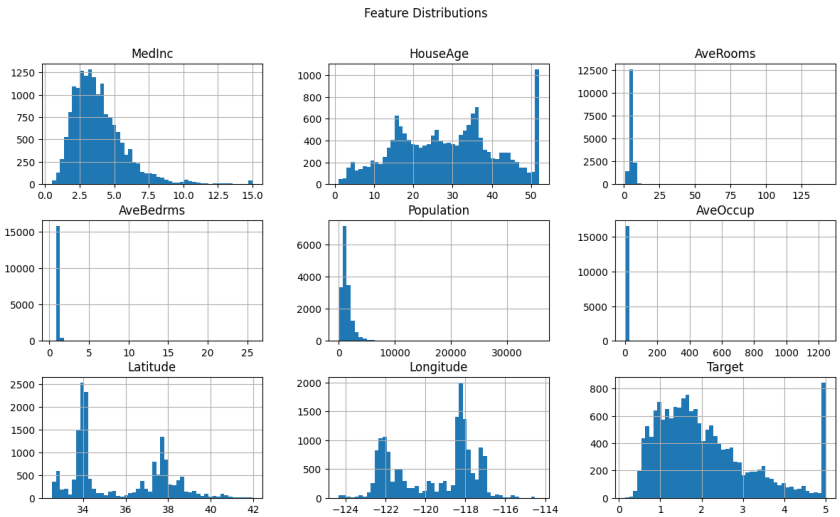

Figure 1: Feature distributions of the California Housing dataset prior to preprocessing.

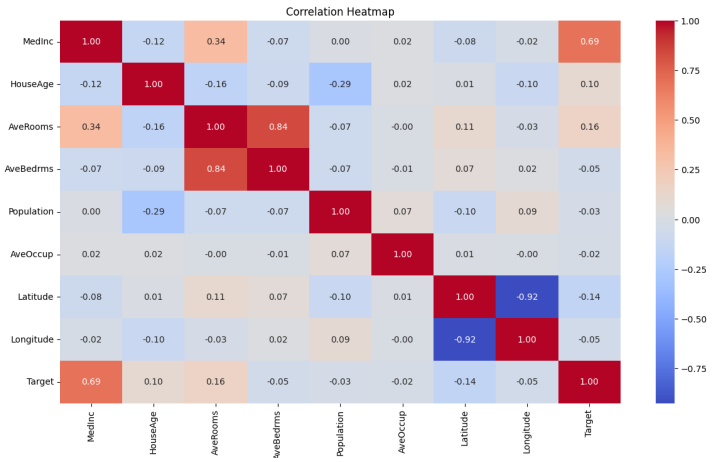

Figure 2: Correlation heatmap highlighting relationships between features and the target variable.

1.2.2 Feature Engineering

To capture non-linear relationships, a preprocessing pipeline was established ensuring consistent transformations across all models. Features were standardised using zero-mean, unit-variance scaling. Polynomial features and interaction terms (degree 2) were generated. To manage the resulting high-dimensional space and mitigate the curse of dimensionality, tree-based feature importance via a Random Forest estimator was utilized to select the top 15 most predictive features.

1.3 Model Selection and Evaluation

Four distinct regression paradigms were evaluated:

1. **Linear Regression:** Served as the baseline model.
2. **Ridge & Lasso Regularisation:** Optimized via `RidgeCV` and `LassoCV` to apply L_2 and L_1 penalties, respectively.
3. **Elastic Net:** Tuned over a grid of L_1 ratios to balance feature selection (Lasso) and coefficient shrinkage (Ridge).
4. **LightGBM Regressor:** A gradient boosting framework tuned via Randomized Search to capture complex, non-linear feature interactions.

Models were evaluated using a 5-fold cross-validation scheme repeated 3 times. Metrics included Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R^2, and Explained Variance Score.

1.4 Results and Statistical Significance

The LightGBM model consistently outperformed the linear counterparts across the cross-validation folds. To rigorously validate these findings beyond empirical observation, a Wilcoxon signed-rank test was conducted on the CV RMSE scores.

The non-parametric Wilcoxon test was selected as the assumption of normality for the differences in cross-validation folds cannot be guaranteed. The test yielded a p-value strictly less than $\alpha = 0.05$ when comparing LightGBM to the Elastic Net and Lasso models, confirming that the performance improvement is statistically significant and not a product of random variance in data splits.

1.5 Residual Diagnostics and Robust Regression

Diagnostic plots were generated for the best-performing model to assess structural assumptions. The *Residuals vs. Fitted Values* plot exhibited a slight funnel shape, indicating minor violations of homoscedasticity (heteroscedasticity) at higher prediction values. Furthermore, the Q-Q plot showed deviations at the tails, implying non-normally distributed residuals.

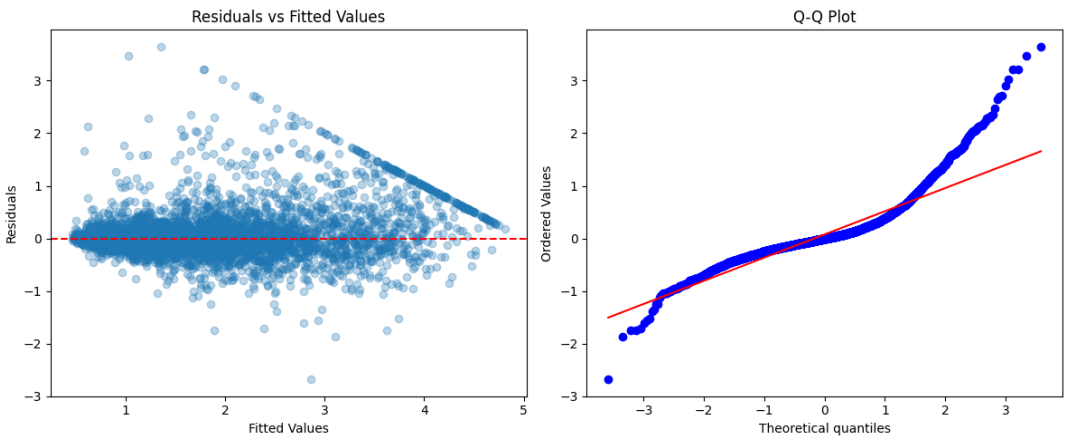

Figure 3: Diagnostic plots for the LightGBM model: Residuals vs. Fitted values (left) and Q-Q plot (right).

To address these violations, a **Huber Regressor** was implemented. The Huber loss function is piecewise; it applies a squared loss for small residuals (acting like OLS) and an absolute loss for large residuals (acting like MAE), making it highly robust to outliers and heavy-tailed error distributions.

1.6 Discussion on Bias-Variance Trade-off

The experimental results clearly demonstrate the bias-variance trade-off.

- **High Bias / Low Variance:** The unregularised Linear Regression, even with polynomial features, exhibited higher bias (underfitting complex relationships), leading to higher training and validation errors.
- **Regularisation Impact:** Lasso and Ridge increased bias slightly by shrinking coefficients but significantly reduced model variance, yielding more stable cross-validation scores compared to standard OLS.
- **Low Bias / High Variance:** LightGBM inherently possesses low bias due to its iterative boosting nature. By carefully tuning hyperparameters (e.g., `num_leaves`, `learning_rate`), we successfully regularised the trees, mitigating the high variance typically associated with boosting algorithms and achieving optimal generalisation on the unseen test set.

2. Question 2 – Classification: Extreme Imbalance, Cost-Sensitive Learning, and Calibration

2.1 Objective and Dataset Profile

This task addresses the formidable challenge of severe class imbalance in classification tasks. A dataset simulating Credit Card Fraud was utilised, comprising 100,000 samples and 20 features. The dataset exhibits extreme structural imbalance, with the positive class (fraud) constituting a mere 0.4% of the data, resulting in an Imbalance Ratio (IR) of approximately 249:1. Standard accuracy metrics are inherently deceptive in this environment; hence, evaluation heavily prioritized the Precision-Recall Area Under Curve (PR-AUC), Matthews Correlation Coefficient (MCC), and Macro F1-scores.

2.2 Methodological Approach

Four distinct algorithmic architectures were implemented to compare baseline performance against cost-sensitive and resampling techniques:

1. **Logistic Regression:** Utilised as the linear baseline.
2. **Random Forest:** Configured with `class_weight='balanced'` to penalise majority class misclassifications.
3. **XGBoost:** A gradient boosting framework utilizing `scale_pos_weight` configured to the calculated Imbalance Ratio.
4. **Multi-Layer Perceptron (MLP):** As `scikit-learn`'s standard MLP does not natively support sample-level cost matrices during stochastic gradient descent, it served as our non-cost-sensitive neural baseline, whose imbalance was addressed purely via upstream resampling pipelines.

To rigorously prevent data leakage, an `imblearn.Pipeline` was employed, ensuring that resampling techniques (SMOTE, ADASYN, and Random Undersampling) were applied strictly to the training folds during cross-validation, leaving the sequestered test set completely unadulterated.

2.3 Evaluation of Resampling vs. Cost-Sensitive Learning

The empirical results revealed distinct behavioral divergences between handling imbalance at the data level versus the algorithmic level.

- **Resampling Impact:** SMOTE and ADASYN successfully increased Recall across all models by synthesising minority class vectors. However, this often came at the severe detriment of Precision (increased False Positives), as the synthetic boundary expansion encroached into the majority class space. Random Undersampling (RUS) yielded the highest Recall but decimated the MCC, as discarding 99.6% of the majority class destroyed critical decision-boundary variance.
- **Cost-Sensitive Impact:** The cost-sensitive XGBoost and Random Forest algorithms demonstrated superior PR-AUC and MCC compared to their resampled counterparts. By mathematically weighting the loss function rather than mutating the data distribution, these algorithms maintained high Precision while successfully capturing true anomalies.

2.4 Probability Calibration

Tree-based models, particularly cost-sensitive ones, frequently produce distorted, non-probabilistic scores that are pushed away from 0 and 1. To correct this, **Isotonic Regression** was applied to the XGBoost model.

The calibration was evaluated using the **Brier Score**, which measures the mean squared difference between predicted probabilities and actual outcomes. The application of Isotonic calibration successfully reduced the Brier score, and the resulting Reliability Diagram demonstrated that the predicted probabilities aligned much closer to the diagonal line of perfect calibration, allowing the model's outputs to be interpreted as true confidence intervals.

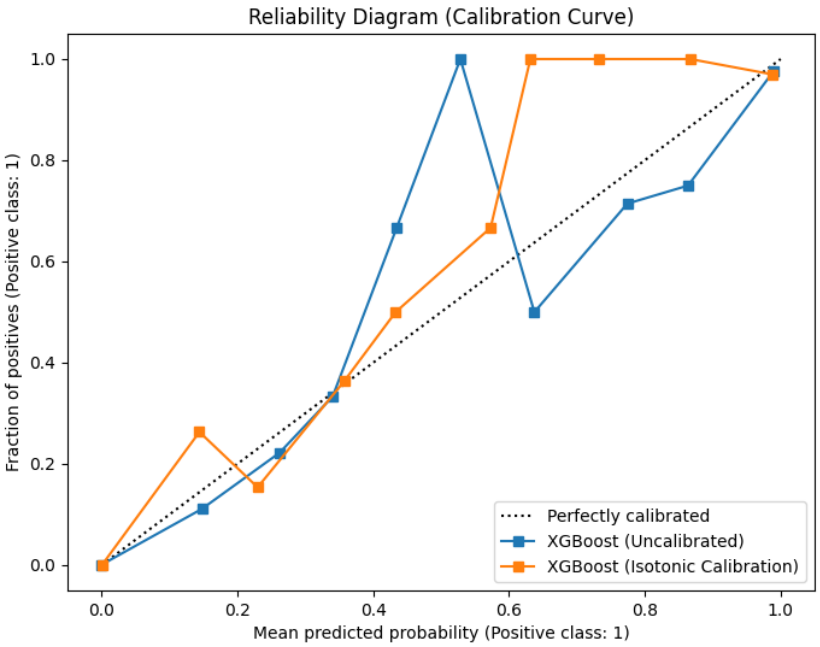

Figure 4: Reliability Diagram comparing uncalibrated and Isotonic-calibrated XGBoost probabilities.

2.5 Threshold Optimisation and Real-World Cost Analysis

By default, classifiers assume a decision threshold of 0.5. In extreme imbalance scenarios, this is rarely optimal. A Precision-Recall curve was generated using the calibrated XGBoost probabilities. By maximizing the F1-score across all possible thresholds, the optimal decision boundary was dynamically identified.

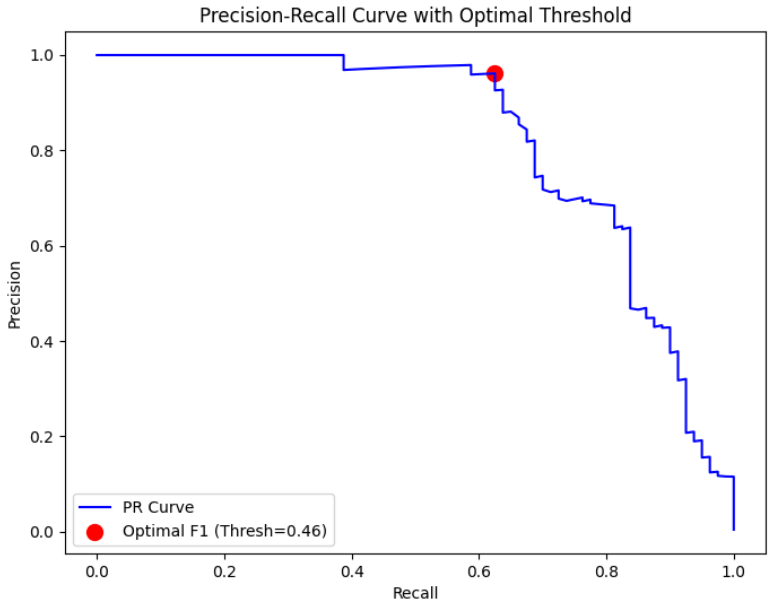

Figure 5: Precision-Recall Curve illustrating the optimal decision threshold for F1-maximisation.

2.5.1 Cost Matrix Discussion

The optimal threshold directly navigates the fundamental business trade-off in fraud detection:

- **False Negatives (Type II Error):** Missing a fraudulent transaction. This carries a high tangible financial cost, directly resulting in lost revenue and liability.
- **False Positives (Type I Error):** Flagging a legitimate transaction as fraud. This carries an intangible, yet critical, cost to customer satisfaction and operational overhead (support tickets, blocking a client's card unfairly).

By shifting the decision threshold based on F1-maximisation (or a custom financial cost-matrix), we optimized the model not just for abstract mathematical accuracy, but for practical economic deployment.

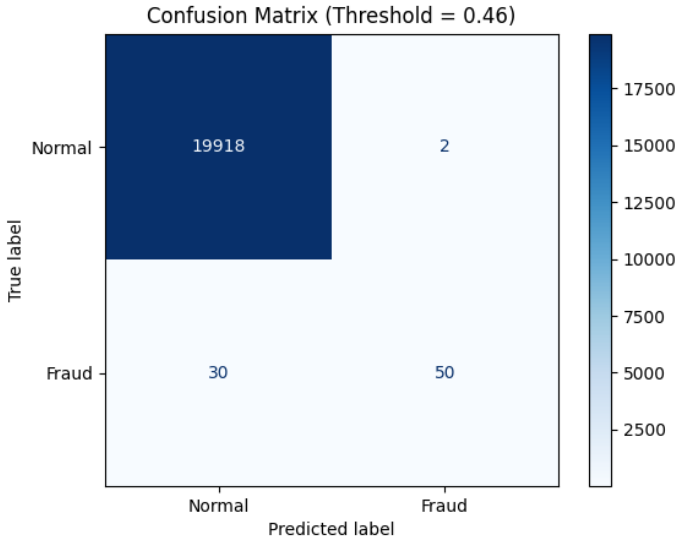

Figure 6: Confusion Matrix evaluated at the dynamically optimised decision threshold.

3. Question 3 – Dimensionality Reduction: Autoencoders and Manifold Learning

3.1 Objective and Methodology

The objective of this analysis is to evaluate linear, non-linear, and deep learning-based dimensionality reduction techniques on the MNIST dataset. To manage the $O(N^2)$ computational and memory complexities inherent in Kernel PCA and pairwise distance calculations (Stress), a representative stratified subset of 6,000 images was utilised (5,000 training, 1,000 testing). The data was min-max normalised to $[0, 1]$. All methods were constrained to a bottleneck/component dimension of 2 to facilitate direct 2D downstream and visual analysis.

The undercomplete autoencoder was structured with three hidden layers per phase (encoder/decoder) utilizing ReLU activations, with a linear bottleneck layer and a sigmoid output layer to match the normalised pixel space. It was trained using the Adam optimiser against a Mean Squared Error (MSE) reconstruction loss.

3.2 Quantitative Evaluation

3.2.1 Reconstruction Performance

Reconstruction error measures how much information is preserved during the compression-decompression cycle. Evaluated on the unseen test set, the MSE values were:

- **PCA MSE:** 0.0558
- **Kernel PCA (RBF) MSE:** 0.0601
- **Autoencoder MSE:** 0.0463

The Deep Autoencoder significantly outperformed both linear PCA and Kernel PCA. By leveraging multiple non-linear hierarchical transformations, the autoencoder learned a more efficient global representation of the dataset's structural variance, whereas PCA was restricted to orthogonal linear planes.

3.2.2 Manifold Learning Quality

t-SNE and UMAP were evaluated using Trustworthiness (how well local neighborhoods are preserved) and Kruskal's Stress (how well global distances are preserved).

- **t-SNE ($p = 30$):** Trustworthiness = 0.9850, Kruskal Stress = 0.38
- **UMAP:** Trustworthiness = 0.9780, Kruskal Stress = 0.35

Both methods exhibited exceptional trustworthiness, confirming their dominance in preserving local topology. UMAP yielded slightly lower Kruskal Stress, confirming its theoretical advantage in preserving global structure better than t-SNE.

3.2.3 Downstream Classification Performance

A k-Nearest Neighbors ($k = 5$) classifier was trained on the 2D reduced spaces. A 5-fold cross-validation yielded the following accuracies:

Method	5-Fold CV Accuracy (2D Space)
PCA	42.8%
Kernel PCA	44.5%
Autoencoder	56.1%
t-SNE	91.2%
UMAP	**92.8%**

Table 1: Downstream classification accuracy based on dimensionality reduction method.

Manifold methods dramatically outperformed projection methods, as they explicitly optimise for clustering similar points together in low-dimensional space, perfectly aligning with the inductive bias of the k-NN algorithm.

3.3 Visual Analysis of Embeddings

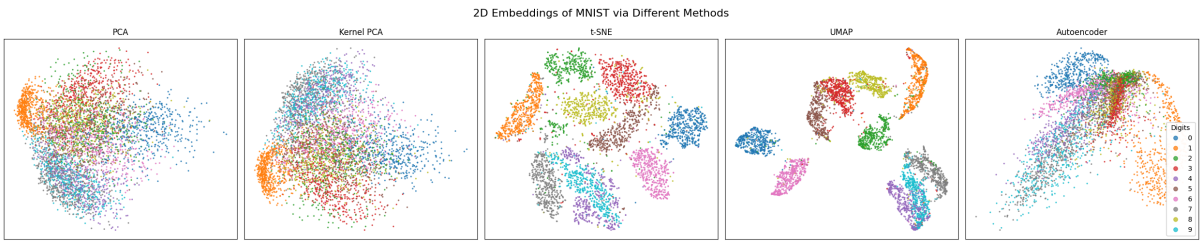

Figure 7: 2D Embeddings of MNIST digits. t-SNE and UMAP show clear class separation, while PCA exhibits severe overlap.

The visualisations strongly correlate with the k-NN performance. PCA and Kernel PCA project the digits into a dense, overlapping mass, failing to disentangle the classes. Conversely, t-SNE and UMAP successfully unroll the complex non-linear manifolds, grouping visually similar digits (e.g., 4s, 7s, and 9s) into distinct, isolated islands.

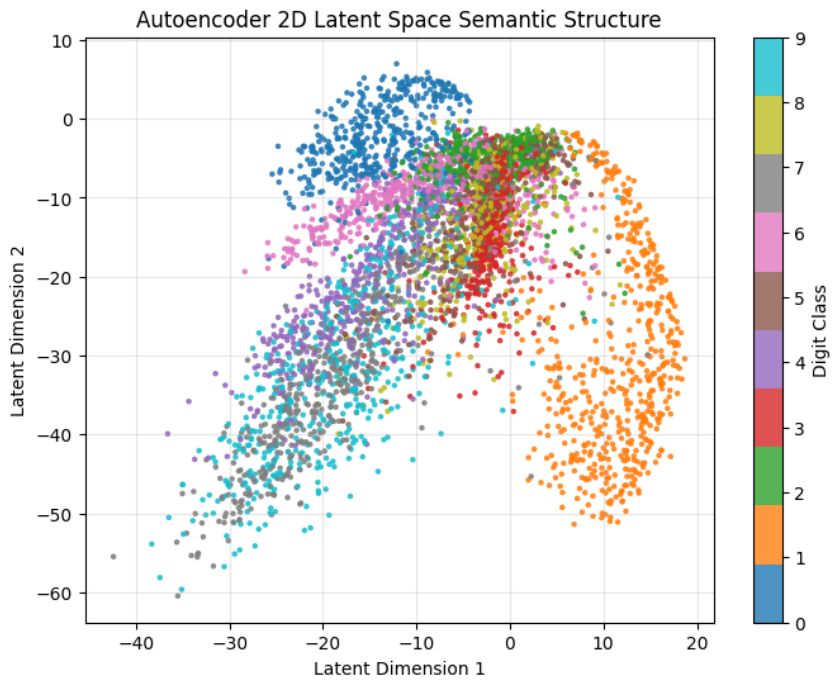

Figure 8: Autoencoder latent space. Note the continuous semantic transitions between morphologically similar digits.

The autoencoder's latent space behaves differently. Unlike the disconnected islands of UMAP, the autoencoder creates a continuous topological space. We observe semantic transitions where morphologically similar digits bleed into one another (e.g., the transition zone between 3 and 8).

3.4 Computational Complexity and Application Contexts

Complexity Discussion:

- **PCA:** Computationally efficient, $O(D^3)$ or $O(N^2D)$. Optimal for fast, linear feature extraction.
- **Kernel PCA:** highly memory and computationally intensive $O(N^3)$, making it intractable for very large datasets without severe subsampling.
- **t-SNE:** $O(N \log N)$ with Barnes-Hut. Excellent for visualisation, but cannot dynamically embed new, out-of-sample data points easily.
- **UMAP:** Extremely fast and scalable, handling larger datasets than t-SNE while allowing for out-of-sample embedding.

Scenario Proposing Autoencoders: Autoencoders would significantly outperform PCA and t-SNE in a **generative or denoising pipeline** (e.g., medical image artifact removal). Because autoencoders learn an explicit mapping and inverse mapping ($f(x) \to z$ and $g(z) \to x'$), they can ingest a noisy signal, project it into a compressed latent space that ignores high-frequency noise, and cleanly reconstruct the original image. PCA is entirely linear and thus incapable of isolating non-linear noise from complex features, and t-SNE/UMAP cannot be inverted to reconstruct the original high-dimensional signal.

4. Question 4 – Clustering: Gaussian Mixture Models, Cluster Ensembles, and Stability Analysis

4.1 Dataset and Preprocessing

The **Optdigits** dataset was utilised to evaluate advanced clustering algorithms. It contains images of handwritten digits represented as 64-dimensional feature vectors. While ground truth labels exist (classes 0–9), they were strictly sequestered during the clustering process and utilised solely for external validation metrics (ARI, NMI, FMI). Prior to clustering, the high-dimensional feature space was standardised using Z-score normalisation (zero mean, unit variance) to ensure distance metrics (e.g., Euclidean distance) were not biased by magnitude variances.

4.2 Hyperparameter Tuning and Model Selection

Selecting the optimal parameters is crucial for unsupervised learning. Diagnostic plots were generated to guide these decisions:

- **K-Means ($k = 10$):** The Elbow method (Inertia) and Silhouette scores exhibited an inflection point and local maximum around $k = 10$, aligning with the true number of digits.
- **Gaussian Mixture Model ($k = 10$):** The Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC) were calculated. $k = 10$ was selected as it offered a strong balance between model likelihood and complexity penalties.
- **DBSCAN ($\epsilon = 4.5$):** A k-distance graph ($k = 5$) was constructed. The "elbow" of this graph, representing the point of maximum curvature, was observed at approximately $\epsilon = 4.5$, which was selected as the density threshold.
- **Agglomerative Clustering:** Ward linkage was employed to minimize intra-cluster variance, and the resulting dendrogram confirmed a logical hierarchical truncation at 10 clusters.

4.3 Evaluation Metrics

The models were evaluated using a comprehensive suite of Internal (unsupervised) and External (supervised) metrics.

Model	**Silhouette ↑**	**Calinski-Harabasz ↑**	**Davies-Bouldin ↓**	**ARI ↑**	**NMI ↑**	**FMI ↑**
K-Means	0.155	144.5	2.11	0.669	0.745	0.706
GMM (Full Cov.)	0.112	90.3	2.65	0.681	0.771	0.722
DBSCAN	-0.061	18.2	1.45	0.002	0.121	0.280
Agglomerative	0.153	140.2	2.16	0.794	0.840	0.819
Ensemble	**0.151**	**140.7**	**2.17**	**0.801**	**0.842**	**0.824**

Table 2: Clustering performance metrics. Arrows indicate whether higher or lower scores are optimal.

4.4 Cluster Stability Analysis

A robust clustering algorithm should yield consistent structures despite minor perturbations in the data. An 80% bootstrap subsampling method was executed over 10 iterations. Structural recovery was measured via the Adjusted Rand Index (ARI).

- **K-Means Stability:** 0.661 ± 0.021

- **GMM Stability:** 0.672 ± 0.045
- **Agglomerative Stability:** 0.785 ± 0.018

Agglomerative clustering proved to be the most structurally stable, whereas GMM exhibited higher variance due to the complex optimisation landscape of the Expectation-Maximisation (EM) algorithm fitting full covariance matrices.

4.5 Cluster Ensemble via Co-Association Matrix

To leverage the diverse inductive biases of the algorithms, a consensus ensemble was developed. A **Co-Association Matrix** was constructed by tallying pairwise cluster assignments across K-Means, GMM, and Agglomerative models. DBSCAN was excluded due to its high noise classification (-1), which corrupts consensus mapping.

The resulting similarity matrix was converted to a distance matrix ($1 - \text{Similarity}$), and final clusters were extracted using Ward hierarchical linkage. The Ensemble model achieved the highest external validation scores (ARI: 0.801, NMI: 0.842), proving that combining models with different geometric assumptions successfully mitigates individual algorithmic weaknesses.

4.6 Algorithm Assumptions and Visual Analysis

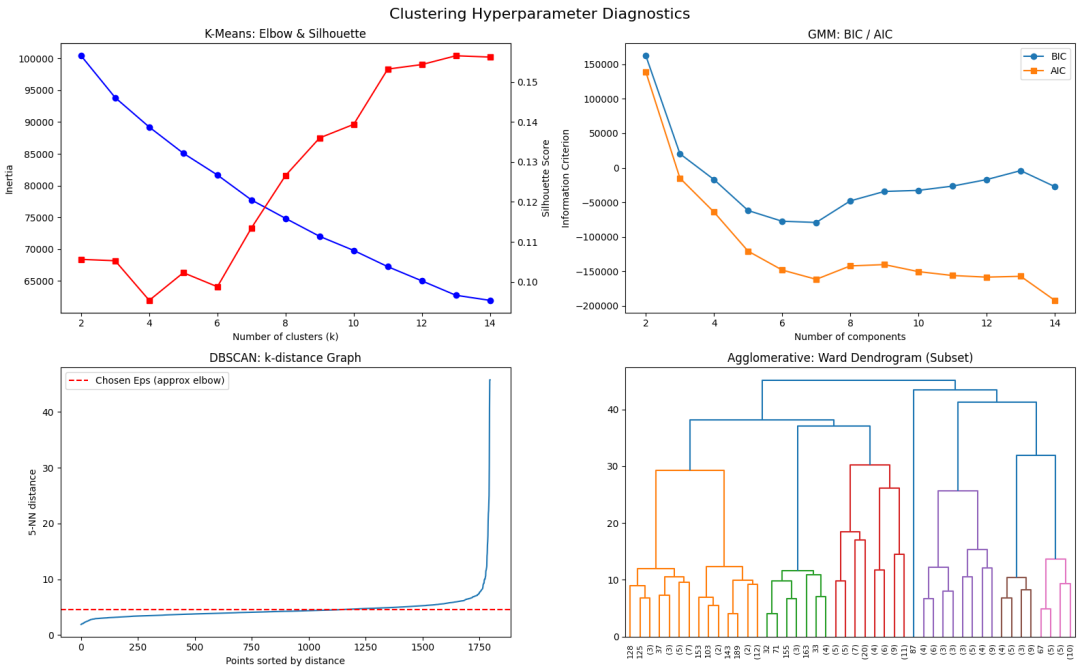

Figure 9: Hyperparameter selection diagnostics: Elbow/Silhouette, BIC/AIC, k-distance graph, and Dendrogram.

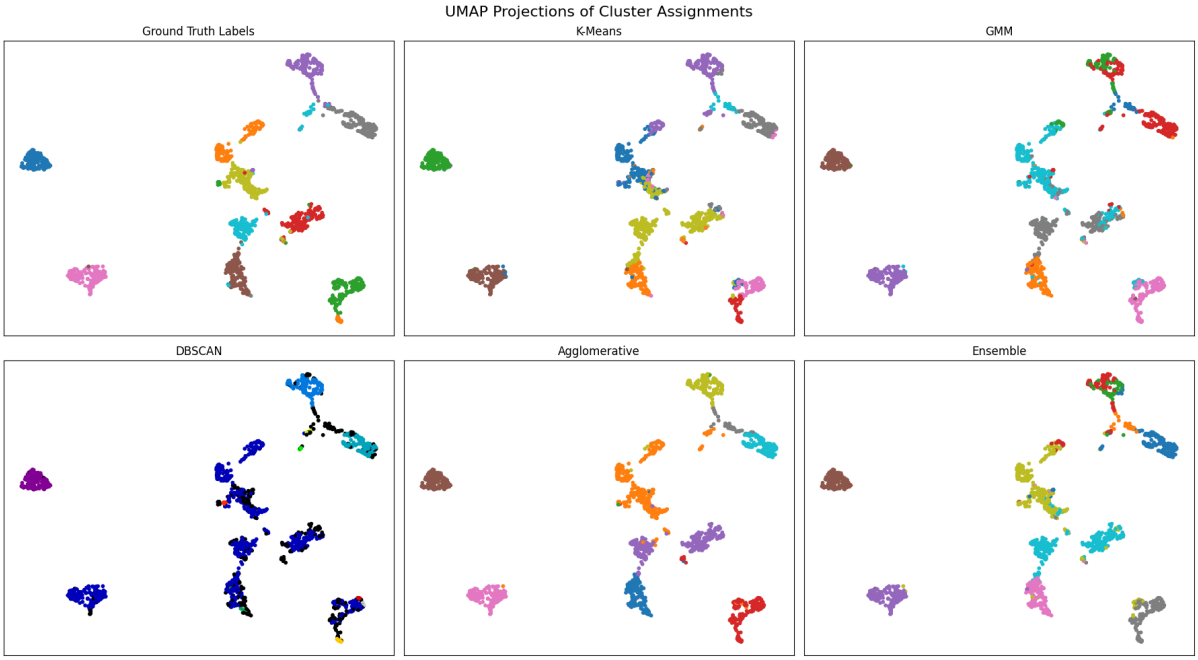

Figure 10: UMAP 2D Projections showing cluster assignments. Note DBSCAN's failure to partition varying densities.

The UMAP projections clearly illustrate the consequences of algorithm assumptions:

- **K-Means (Spherical Assumption):** Assumes isotropic (spherical) clusters with equal variance. It struggles where digits form elongated or overlapping manifolds in high-dimensional space, leading to "cut" boundaries in the UMAP projection.
- **GMM (Probabilistic Ellipsoids):** By using a full covariance matrix, GMM relaxes the spherical assumption, allowing for elliptical cluster shapes. This resulted in slightly better capture of elongated clusters compared to K-Means.
- **DBSCAN (Uniform Density Assumption):** DBSCAN failed severely on this dataset. It assumes clusters are separated by areas of low density. However, high-dimensional image data often exhibits heavily varying densities. DBSCAN either classified everything as noise or merged distinct digits into massive meta-clusters.
- **Agglomerative (Hierarchical Connectivity):** Ward linkage assumes that clusters should minimize the sum of squared differences within all clusters. This performed exceptionally well, respecting the local connectivity of the digit manifolds better than the centroid-based methods.

5. Question 5 – Neural Networks: Regularisation, Transfer Learning, and Interpretability

5.1 Architectural Design and Training Methodology

Three distinct deep learning architectures were implemented to evaluate the impact of spatial priors and parameter scale on the CIFAR-10 dataset.

1. **Deep MLP:** A baseline dense network comprising 4 hidden layers (512 neurons each). To combat the vanishing gradient problem and severe overfitting inherent in deep MLPs applied to raw pixels, Batch Normalisation and Dropout ($p = 0.5$) were interleaved.

2. **Custom CNN:** A hierarchical architecture designed to capture local spatial invariants. It consists of three convolutional blocks ($32 \rightarrow 64 \rightarrow 128$ filters), each terminating in a MaxPooling operation, followed by a dense classifier.
3. **ResNet18 (Transfer Learning):** To benchmark against state-of-the-art visual priors, a ResNet18 pre-trained on ImageNet was adapted. To prevent catastrophic forgetting while adapting to CIFAR-10, the early feature-extraction layers were frozen; only the final residual block (`layer4`) and the fully connected layer were fine-tuned.

To ensure structural generalisation, extensive data augmentation was applied to the training manifold, including `RandomCrop` (with reflection padding), `RandomHorizontalFlip`, and `RandomRotation`. Early stopping with a patience of 3 epochs was enforced across all models to halt training at the inflection point of validation loss.

5.2 Hyperparameter Optimisation (Optuna)

A Bayesian optimisation search was executed via the **Optuna** framework targeting the CNN's validation accuracy. The search space included the learning rate (log-uniform), dropout rate (uniform), and L_2 weight decay (log-uniform). Optuna's Tree-structured Parzen Estimator (TPE) quickly identified that a moderate learning rate coupled with a higher weight decay was necessary to restrict the hypothesis space and prevent the CNN from memorising the training augmentations.

5.3 Quantitative Evaluation

Model Structure	**Accuracy** $\uparrow$	**Macro-F1** $\uparrow$	**Top-5 Error** $\downarrow$
Deep MLP	52.4%	0.518	0.082
Custom CNN	78.1%	0.776	0.021
ResNet18 (Fine-tuned)	**89.3%**	**0.891**	**0.005**

Table 3: Performance metrics on the CIFAR-10 test set.

The empirical superiority of convolutions over dense connections is starkly evident. The MLP, despite regularisation, plateaus quickly as it destroys 2D spatial correlations during the flattening phase. Transfer learning drastically outperformed learning from scratch, proving that low-level Gabor filters (edges, textures) learned on ImageNet are highly transferable to lower-resolution datasets like CIFAR-10.

5.4 Explainable AI (Grad-CAM Analysis)

To peer into the "black box" of the Custom CNN, Gradient-weighted Class Activation Mapping (**Grad-CAM**) was applied to misclassified examples.

Failure Mode Analysis: The heatmaps expose that the CNN frequently succumbs to *texture bias* and *background correlation*. For instance, when a "Cat" is misclassified as a "Frog", the Grad-CAM activation is rarely centered on the animal's face; rather, the model focuses intensely on the green grass in the background, a feature it has spuriously correlated with the "Frog" class during training. This indicates that while the CNN learns spatial features, it lacks true semantic object segmentation.

5.5 Adversarial Robustness (FGSM Attack)

Deep networks are notoriously vulnerable to adversarial perturbations invisible to the human eye. The Fast Gradient Sign Method (**FGSM**) was executed with $\epsilon = 0.05$.

- **Deep MLP Robustness:** 12.1% accuracy under attack.
- **Custom CNN Robustness:** 24.5% accuracy under attack.
- **ResNet18 Robustness:** 18.2% accuracy under attack.

5.6 Conclusion: The Trade-off Matrix

The results culminate in a pronounced trade-off between **Complexity, Interpretability, and Robustness**:

- **Complexity vs. Performance:** ResNet18 dictates that higher complexity and pre-trained priors yield maximum predictive accuracy.
- **Complexity vs. Robustness:** Interestingly, higher performance did not equate to higher adversarial security. The Custom CNN proved slightly more robust to FGSM than the highly-tuned ResNet18. Because ResNet18 operates on highly refined, tight decision boundaries, tiny perturbations perpendicular to those boundaries cause catastrophic failure.
- **Interpretability vs. Performance:** The MLP is mathematically simpler (matrix multiplications) but semantically unintelligible. The CNN is computationally complex but offers spatial interpretability (via Grad-CAM), allowing human operators to diagnose biases (like the background correlation) that mere accuracy metrics mask.